Bauhaus-Universität Weimar
Faculty of Media
Degree Programme Computer Science for Digital Media

Generative Agents School: The Impact of Thuringian Government Programs on Generative Students

Social Simulacra

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Abstract

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Chapter 1

Introduction

Education represents one of the primary pillars of human society, serving as a vital mechanism for cultural transmission, social reproduction, and civic preparation [Dew16; BP70]. Since antiquity, older generations have transmitted knowledge, skills, and cultural norms to younger generations, enabling societies to build upon accumulated experience and advance institutionally. Modern states formalize this process through public schooling, establishing standardized environments wherein children acquire the necessary competencies to participate in democratic life [Dew16].

However, formal education is inherently value-laden; decisions regarding what knowledge is essential and how classrooms should be governed are fundamental political questions [App04]. In democratic systems such as Germany, state governments determine educational policies, curriculum frameworks, and school structures. Political parties hold sharply contrasting visions of educational priorities:

- **Die Linke** emphasizes social equality, advocating for comprehensive, non-selective schooling and the elimination of educational barriers.
- **SPD** focuses on equal opportunity, educational mobility, and state-supported foundational skills.
- **Bündnis 90/Die Grünen** prioritizes climate education, social inclusion, modern pedagogy, and democratic participation.
- **CDU** maintains a traditional view centered on performance standards, discipline, and the classic multi-track school system.
- **AfD** advocates for traditional performance ideals, national cultural preservation, and strict ideological neutrality in the classroom.

In practice, democratic coalition building forces political parties to compromise, meaning that pure ideological platforms are rarely observed in isolation within real-world classrooms. Evaluating how these uncompromised party programs independently affect student well-being, social dynamics, and developmental trajectories presents a methodological challenge.

To overcome real-world policy blurring and isolate party-specific educational visions, this project constructs a simulated educational environment using the generative agents architecture introduced by Park et al. [Par+23]. By instantiating autonomous LLM-driven agents within a controlled sandbox environment, we translate the distinct educational programs of *Die Linke*, *SPD*, *Die Grünen*, *CDU*, and *AfD* into explicit school rules and classroom curricula. Holding the baseline environment constant, we fork a single initial state into five parallel timelines, simulating a full school day under each party's distinct regime. To evaluate the impact of each party framework, we conduct structured interviews with student agents before the simulation fork and after the completion of the simulated school day, probing variations in their emotional state, peer relationships, and academic perceptions.

Chapter 2

Related Work

Chapter 3

Setting the Environment

Chapter 4

Agent Interviews

4.1 Evaluation Framework for Personality Shifts

Personality evaluation in the InterviewSim framework relies on eliciting qualitative open-ended responses from agents and converting these responses into Big Five (OCEAN) trait ratings using an LLM judge [Li+26]. This section outlines the protocol to measure how specific school curricula and regulations alter student and teacher personality traits across five timeline forks.

4.1.1 Interview Methodology Grounding

To detect personality shifts, the evaluation protocol follows a four-step pipeline [Li+26]. First, we conduct structured natural language interviews with the four students and one teacher at baseline prior to the fork and at the end of the school day across all five timelines. Second, interview questions target specific behavioral categories, focusing on psychological traits along with motivations and values. Third, an LLM judge evaluates each agent response set across five independent runs per trait with a temperature above zero. Applying majority voting via the statistical mode across these runs eliminates measurement variance. Fourth, categorical outputs map to an ordinal scale to quantify trajectory changes across timelines.

4.1.2 Big Five Interview Question Set

The interview protocol combines general trait probes with scenario-based school contexts to extract authentic behavioral tendencies [Li+26]. The same question set is administered in both pre-fork and post-simulation interviews.

Openness to Experience

Openness questions evaluate intellectual curiosity, adaptability to new rules or curricula, and creative expression.

1. How do you feel when forced to adapt to a completely new set of rules or a new learning method?
2. What was the most unexpected or challenging idea presented in class today, and how did you engage with it?
3. If given full freedom, how would you prefer to spend your time during lessons or free periods?

Conscientiousness

Conscientiousness questions target orderliness, duty, rule adherence, and self-discipline.

1. How important is it for you to follow rules strictly, even when you disagree with them?
2. How did you handle your tasks, assignments, or teaching responsibilities during the school day?
3. What goes through your mind when someone disrupts the structure or order of the classroom?

Extraversion

Extraversion questions measure energy, social engagement, assertiveness, and enthusiasm.

1. Do you feel energized or exhausted after spending a day actively communicating with peers or students?
2. How active were you in discussions during break times or class conversations today?
3. When a group disagreement arises in school, do you step forward to voice your opinion or stay back?

Agreeableness

Agreeableness probes target empathy, cooperation, trust, and compliance versus antagonism.

1. How do you usually react when someone questions your authority or challenges your views?
2. How did you interact with your fellow students or teacher today when someone needed help or made a mistake?
3. Do you prioritize maintaining harmony with others, or pushing for what you personally think is right?

Neuroticism

Neuroticism questions measure emotional stability, anxiety, stress response, and reactivity.

1. How do you handle stressful or tense situations in an academic environment?
2. Did you feel anxious, frustrated, or overwhelmed at any point during the school day today, and why?
3. When things do not go according to plan in class, how quickly do you recover your composure?

4.1.3 Analytical Framework for Personality Shift

The quantitative measurement pipeline converts interview responses into formal shift metrics.

LLM-as-a-Judge Evaluation

The complete interview transcript of an agent is provided to an LLM judge, such as GPT-4o [Li+26]. The judge evaluates each Big Five trait individually, assigning a level of *Low*, *Neutral*, or *High*. Following standard protocols, the judge receives specific behavioral indicators for each level before issuing a structured tag rating alongside a written justification.

Majority Voting and Quantification

To guarantee statistical stability, the judge evaluates each transcript five times per trait [Li+26]. Equation (1) defines the final trait level as the statistical mode of these ratings:

$$\text{Trait Level}(t) = \text{Mode}\Big(\{r_1, r_2, r_3, r_4, r_5\}\Big), \quad r_i \in \{\text{Low}, \text{Neutral}, \text{High}\}. \tag{4.1}$$

Categorical ratings are subsequently mapped to ordinal numerical values: $m(\text{Low}) = 1.00$, $m(\text{Neutral}) = 2.00$, and $m(\text{High}) = 3.00$.

Shift and Alignment Metrics

For any given timeline $k \in \{1, 2, 3, 4, 5\}$, Equation (2) calculates the trait shift Δt_k from the baseline interview t_{base} to the post-simulation interview $t_{\text{post},k}$:

$$\Delta t_k = m(t_{\text{post},k}) - m(t_{\text{base}}). \tag{4.2}$$

Equation (3) defines the overall personality alignment between baseline and post-simulation agent states across all five OCEAN traits:

$$\text{Alignment}_k = 1.00 - \frac{\sum_{t \in \text{OCEAN}} |m(t_{\text{base}}) - m(t_{\text{post},k})|}{10.00}. \tag{4.3}$$

Cross-Timeline Comparison

Table [**tab:trait_shifts**] demonstrates the format for mapping personality divergence across timelines, facilitating direct attribution of trait changes to specific educational frameworks.

4.2 Cross-Timeline Big Five Shift Matrices

This section details the empirical Big Five personality trait shifts across all political party timeline forks: *Die Linke* (LINKE), Social Democratic Party (SPD), Alliance 90/The Greens (GRUENE), Christian Democratic Union (CDU), and Alternative for Germany (AFD). Tables **??** and **??** summarize the ordinal trait values (1.00 = Low, 2.00 = Neutral, 3.00 = High) for each student and teacher agent before (Baseline) and after the simulated school day.

Table 4.1: Big Five trait shifts for student and teacher agents across political party timelines (O: Openness, C: Conscientiousness, E: Extraversion, A: Agreeableness, N: Neuroticism).

Role	Trait	Base	LINKE	SPD	GRUENE	CDU	AFD	Max Δt
Teacher								
Katharina	(O)	3.00	3.00	3.00	3.00	2.00	1.00	-2.00
	(C)	3.00	2.00	3.00	2.00	3.00	3.00	-1.00
	(E)	2.00	3.00	2.00	3.00	2.00	1.00	$+1.00/-1.00$
	(A)	3.00	3.00	3.00	3.00	2.00	1.00	-2.00
	(N)	1.00	1.00	1.00	1.00	2.00	3.00	$+2.00$
Students								
Mohammed	(O)	3.00	3.00	3.00	3.00	2.00	1.00	-2.00
	(C)	2.00	2.00	2.00	2.00	3.00	3.00	$+1.00$
	(E)	3.00	3.00	3.00	3.00	2.00	1.00	-2.00
	(A)	2.00	3.00	2.00	3.00	2.00	1.00	$+1.00/-1.00$
	(N)	1.00	1.00	1.00	1.00	2.00	3.00	$+2.00$
Aylin	(O)	3.00	3.00	3.00	3.00	2.00	1.00	-2.00
	(C)	2.00	2.00	2.00	2.00	3.00	3.00	$+1.00$
	(E)	2.00	3.00	2.00	2.00	2.00	1.00	$+1.00/-1.00$
	(A)	3.00	3.00	3.00	3.00	2.00	1.00	-2.00
	(N)	2.00	1.00	2.00	1.00	2.00	3.00	$+1.00/-1.00$
Lukas	(O)	2.00	2.00	2.00	2.00	2.00	1.00	-1.00
	(C)	3.00	2.00	3.00	2.00	3.00	3.00	-1.00
	(E)	2.00	2.00	2.00	2.00	2.00	2.00	0.00
	(A)	2.00	2.00	2.00	2.00	2.00	2.00	0.00
	(N)	1.00	1.00	1.00	1.00	1.00	2.00	$+1.00$
Sofia	(O)	3.00	3.00	3.00	3.00	2.00	2.00	-1.00
	(C)	1.00	1.00	2.00	2.00	2.00	2.00	$+1.00$
	(E)	3.00	3.00	3.00	3.00	2.00	1.00	-2.00
	(A)	3.00	3.00	3.00	3.00	2.00	1.00	-2.00
	(N)	2.00	1.00	2.00	1.00	2.00	3.00	$+1.00/-1.00$

Appendix A

My First Appendix

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Bibliography

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